INCH-POUND

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MILITARY SPECIFICATION

STUD WELDING SYSTEMS, DC, INTEGRAL POWER SOURCE AND CONTROL UNIT, ELECTRIC ARC AND CAPACITOR DISCHARGE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers portable electric arc and capacitor discharge stud welding systems consisting of an integral direct current (DC) power source, timer controls, stud gun(s), and cables. Stud welding units with independent power supplies and control units are covered by MIL-W-8011O. All requirements specified in this document are the minimum requirements of the Government and are not intended to limit the manufacturer in machine design or performance.

1.2 Classification. The stud welding systems shall be of the following types and sizes. The type and size to be furnished shall be as specified (see 6.2).

Type II - Capacitor discharge (stored energy) system consisting of an integral power source, timer controls, stud gun, and cables.

Size	Stud diameter range
1	12 gauge through l/4-inch
2	12 gauge through 5/16-inch
3	l/8-inch through 3/8-inch

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3431

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Type VII - Electric arc welding system consisting of an integral DC power source with timer controls, stud gun(s), and cables.

Size	Stud diameter range
1	10 gauge through l/2-inch (non-regulated power supply)
2*	10 gauge through 5/8-inch (regulated power supply)
3*	10 gauge through 3/4-inch (regulated power supply)
4*	10 gauge through 7/8inch (regulated power supply)
5*	5/16 inch through l-inch (regulated power supply)
6	10 gauge through l-1/4-inch (regulated power supply)

(NOTE: The asterisk (*) indicates a welding system which is available with either one or two stud guns.)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these domments are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

MILITARY

MS-51336 - Lunette, Coupler, Drawbar, Ring.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Z49.1 - Safety in Welding and Cutting. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

AMERICAN WELDING SOCIETY

AWS C5.4 - Recommended Practices for Stud Welding.

(Application for copies should be addressed to the American Welding Society, Inc., ATTN: Order Dept., 550 N. W. LeJuene Road, Miami, FL 33135-1040.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete system shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

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3.2 Design. The stud welding system shall be new (not a prototype) and one of the manufacturer's current production models capable of operation in accordance with the requirements herein. The system shall weld studs to the surface of ferrous and nonferrous materials. Welds shall be made without deterioration of the stud or base metal. Stud sizes specified for each system shall be the weld base diameter of the stud. The welding system shall withstand rough handling encountered in construction and fabrication work. The design shall provide for safe operation. The system shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Reclaimed materials, The welding system may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the system. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.2 Energy efficiency. The welding system and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry as provided on the manufacturer's standard commercial product.

3.2.3 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the welding system, whether or not specifically required herein, shall comply to all requirements of OSHA 29 CFR 1910 and ANSI Z49.1. If a conflict arises between 29 CFR 1910 and ANSI 249.1, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Safety devices normally furnished as standard on the manufacturer's commercial equipment supplied to the commercial market shall be provided for all other parts of the welding system that present safety hazards.

3.2.4 Mercury restriction. The system shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.5 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the system.

3.2.6 Environmental protection. During the construction of the welding system, the manufacturer shall comply with all applicable Federal, state, and local statutes which regulate the use of materials hazardous to the ecological system in effect at the point of manufacture on the date of the contract. During the operation of the welding system at its final location, it shall not emit materials hazardous to the environment as prohibited by Federal statutes.

3.2.7 Interchangeability. All parts subject to replacement shall be manufactured to definite dimensions and tolerances. Replacement parts shall be interchangeable without requiring modification. Accessories normally supplied as standard items shall also be considered replacement par parts Replacement parts shall be domestically available.

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3.3 Construction". The system shall be constructed of parts which are new, without defects, and free of repairs. The system shall operate at its maximum rating and capacity without permanent damage or failure.

3.3.1 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.2 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.3 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

3.3.4 Painting. Unless otherwise specified (see 6.2), the welding system shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.5 Threads. All threaded parts used on the welding system and its related attachments and accessories shall conform to FED-STD-H28 and the applicable 'Detailed Standard" section referenced therein.

3.4 Type II capacitor discharge system. Type II capacitor discharge systems shall consist of a solid state capacitor discharge (stored energy) power source, power cord, ventilation, stud gun with connectors, control unit, and weld, control, and ground cables. Type II systems shall be initial contact, drawn arc, or initial gap capacitor discharge systems as specified (see 6.2). Voltage control, power discharge control, straight or reverse polarity, an on-off switch with pilot light, and positive and negative control receptacles shall be provided. Drawn arc systems, when specified, shall also be provided with arc timing control. Each system power source shall be capable of meeting the stud welding rates given; however, actual weld rates will be limited by operator capability.

3.4.1 Type II, size 1. Type II, size 1 systems shall be capable of welding studs 12 gauge through l/4-inch diameter. The stud welding rate shall be not less than 8 per minute for l/4-inch diameter studs. Unless otherwise specified (see 6.2), type 11, size 1 systems shall operate from 120 volt, 20 ampere, 60 Hertz (Hz), single-phase alternating current (AC) input power.

3.4.2 Type II, size 2. Type II, size 2 systems shall be capable of welding studs 12 gauge through 5/16-inch in diameter. The stud welding rate shall be not less than 10 per minute for 5/16-inch diameter studs, Unless otherwise specified (see 6.2), type II, size 2 systems shall operate from 120 volt, 20 ampere, 60 Hz, single-phase AC input power.

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3.4.3 Type II, size 3. Type II, size 3 systems shall be capable of welding studs 1/8-inch through 3/8-inch in diameter. The stud welding rate Shall be not less than 10 per minute for 3/8-inch diameter studs. Unless otherwise specified (see 6.2), type II, size 3 systems shall operate from 120 volt, 30 ampere, 60 Hz, single-phase AC input power.

3.4.4 Capacitor discharge safety devices. Type II system control units shall automatically discharge the power capacitors within 30 seconds after the control unit is de-energized to eliminate the possibility of shock or arcing stored energy in the capacitor bank. When specified (see 6.2), a protective circuit shall also be provided to prevent the capacitor voltage from being applied to the stud gun until the gun is, securely seated on the workpiece to prevent shock to the operator.

3.5 Type VII electric arc systems. Type VII electric arc systems shall consist of a solid state, integral, DC, transformer-rectifier power source, ventilation, stud gun(s) with connecting devioes, and cables. Each system shall also be provided with a built-in control unit, including adjustable timing and current controls, and a power on-off switch with pilot light. Current and time controls shall be adjustable through a range which will allow the system to meet the requirements herein. Each system power source shall be capable of meeting the stud welding rates given; however, actual weld rates will be limited by operator capability.

3.5.1 Type VII, size 1. The type VII, size 1 system shall be capable of welding studs 10 gauge through l/2-inch in diameter. The stud welding rate shall be not less than 14 per minute for 1/4-inch diameter studs, 12 per minute for 5/16 inch diameter studs, 10 per minute for 3/8-inch diameter studs, and 5 per minute for 1/2 inch diameter studs. The maximum output current of the type VII, size 1 system shall be not less than 800 amperes. Unless otherwise specified (see 6.2), type VII, size 1 systems shall operate from 230/460 volt, 60 Hz, 3-phase AC input power and shall be initially wired for operation on 460 volts.

3.5.2 Type VII, size 2. The type VII, size 2 system shall be capable of welding studs 10 gauge through 5/8-inch in diameter. The system shall have automatically-regulated weld current output to compensate for incoming line fluctuation. The stud welding rate shall be not less than 30 per minute for 3/8-inch diameter studs, 12 per minute for l/2-inch diameter studs, and 6 per minute for 5/8-inch diameter studs. The maximum output current of the type VII, size 2 system shall be not less than 1200 amperes. Unless otherwise specified (see 6.2), type VII, size 2 systems shall operate from 230/460 volt, 60 Hz, 3-phase AC input power and shall be initially wired for operation on 460 volts.

3.5.2.1 Type VII, size 2 heavy duty gun. When specified (see 6.2), the type VII, size 2 system shall be provided with a heavy duty gun to weld studs 12 gauge through l-inch in diameter. The heavy duty gun shall have a stud welding rate of not less than 20 per minute l/2-inch diameter studs, 18 per minute 5/8-inch diameter studs, and 6 per minute 3/4-inch diameter studs.

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3.5.2.2 Type VII, size 2 dual guns. When specified (see 6.2), the type VII, size 2 system shall be provided with two stud guns (either standard or heavy duty as specified in 3.5.2.1), complete with dual controls and cables. The dual gun system shall have the same design and performance characteristics as the single (standard or heavy duty) gun system.

3.5.3 Type VII, size 3. The type VII, size 3 system shall be capable of welding studs 10 gauge through 3/4-inch in diameter. The size 3 system shall have automatically-regulated weld current output to compensate for incoming line fluctuation. The welding rate shall be not less than 30 per minute for l/2-inch diameter studs, 18 per minute for 5/8-inch diameter studs, and 10 per minute for 3/4-inch diameter studs. The maximum output current for type VII, size 3 systems shall be not less than 1600 amperes. Unless otherwise specified (see 6.2), type VII, size 3 systems shall operate from 230/460 volt, 60 Hz, 3-phase AC input power and shall be initially wired for operation on 460 volts .

3.5.3.1 Type VII, size 3 dual guns. When specified (see 6.2), the type VII, size 3 system shall be provided with two stud guns complete with dual controls and cables. The dual gun system shall have the same design and performance characteristics as the single gun system.

3.5.4 Type VII, size 4. The type VII, size 4 system shall be capable of welding studs 10 gauge through 7/8-inch in diameter. The size 4 system shall have automatically-regulated weld current output to compensate for incoming line fluctuation. The welding rate shall be not less than 12 per minute for 3/4-inch diameter studs and 10 per minute for 7/8inch diameter studs. The maximum output current for type VII, size 4 systems shall be not less than 1800 amperes. Unless otherwise specified (see 6.2), type VII, size 4 systems shall operate from 230/460 volt, 60 Hz, 3-phase AC input power and shall be initially wired for operation on 460 volts.

3.5.4.1 Type VII, size 4 dual guns. When specified (see 6.2), the type VII, size 4 system shall be provided with two stud guns complete with dual controls and cables. The dual gun system shall have the same design and performance characteristics as the single gun system.

3.5.5 Type VII, size 5. The type VII, size 5 system shall be capable of welding studs 5/16 inch through l-inch in diameter. The size 5 system shall have automatically-regulated weld current output to compensate for incoming line fluctuation. The welding rate shall be not less than 18 per minute for 3/4-inch diameter studs, 15 per minute for 7/8-inch diameter studs, and 10 per minute for l-inch diameter studs. The maximum output current for type VII, size 5 systems shall be not less than 2400 amperes. Unless otherwise specified (see 6.2), type VII, size 5 systems shall operate from 230/460 volt, 60 Hz, 3-phase AC input power and shall be wired for operation on 460 volts.

3.5.5.1 Type VII, size 5 dual guns. Uhen specified (see 6.2), the type VII, size 5 system shall be provided with two stud guns complete with dual controls and cables. The dual gun system shall have the same design and performance characteristics as the single gun system.

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3.5.6 Type VII, size 6. The type VII, size 6 system shall be capable of welding studs 10 gauge through l-1/4-inches in diameter. The size 6 system shall have automatically-regulated weld current output to compensate for incoming line fluctuation. The welding rate shall be not less than 18 per minute for 3/4-inch diameter studs, 15 per minute for 7/8-inch diameter studs, 10 per minute for l-inch diameter studs, and 10 per minute for l-1/4-inch diameter studs. The maximum output current for type VII, size 6 systems shall be not less than 2500 amperes. Unless otherwise specified (see 6.2), type VII, size 6 systems shall operate from 230/460 volt, 60 Hz, 3-phase AC input power and shall be initially wired for operation on 460 volts.

3.5.7. Rectifier. A full wave semiconductor rectifier shall be provided on type VII systems. All components shall be of suitable size and design for the intended service. Surge suppression protection shall be provided with capacity to reduce any transient voltage peaks which would cause component failure. Components subject to failure shall be easily removable to permit replacement.

3.5.8 Transformer. The transformer on type VII systems shall produce a continuous current which, when rectified, shall have a direct current rating as necessary to meet the requirements herein. The transformer shall be of the laminated core or shell type. The transformer secondary shall be insulated from the primary and shall not be of the oil-filled type. Coils shall be provided with insulation between the windings, leads, core, and enclosure. coils shall be protected from moisture. The transformer shall be rigidly mounted within the enclosure, All terminals shall be permanently marked.

3.5.9 Memory analyzer board. When specified (see 6.2), type VII systems shall be provided with a memory analyzer board to record into memory the actual weld cycle parameters. The memory analyzer will allow the operator to check for deviations between programmed and actual weld parameters after the completion of a weld cycle.

3.6 Enclosure. Each welding system shall be encased in a ventilated, dripproof metal enclosure with the strength and rigidity necessary to resist the abuse encountered in a stud welding environment without increasing the risk of fire, electric shock, or operator injury. Metal parts shall be protected from corrosion. Access openings shall be easily removable for inspection, cleaning, and servicing. All openings shall be protected to prevent the entrance of trash or rodents.

3.7 Ventilation system. Type II and type VII, size 1 systems shall be cooled either naturally by air flow enclosure louvers or by a forced air cooling system. Type VII, size 2 through size 6 systems shall have a forced air cooling system to minimize internal enclosure heating of the system and allow all components to meet the performance requirements herein. Forced air cooling shall be provided by a totally enclosed, prelubricated, ball bearing fan motor(s). The fan assembly shall be mounted to withstand vibration. Fan blades shall be of a material and design that will prevent fatigue failure under conditions for which the system is designed.

3.8 Insulation. The class of insulation system used shall permit continuous transformer operation and performance.

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3.9 Automatic failure protection. The primary line voltage feeding the system shall be automatically removed in the event of an overload, excessive temperature rise, excessive weld cycle time, or component failure. All control circuits shall be protected by fuses or circuit breakers. All fuses and circuit breakers shall be accessible for replacement or manual resetting.

3.10 Portability. Unless otherwise specified (see 6.2), each type II system shall be manually portable. Unless otherwise specified (see 6.2), each type VII system shall be manually portable or base mounted and provided with means for rolling on wheels; lifting at a designated point by crane or hoisting mechanism equipped with a hook, shackle, or clevis; or lifting by fork lift truck. The lifting device, when provided, shall be firmly and rigidly attached to the enclosure, capable of supporting 2-1/2 times the system weight.

3.11 Incoming AC power generating set. When specified (see 6.2), a diesel engine driven AC generating set shall be provided to supply incoming power to the stud welding system for remote job site welding where commercially supplied AC power is limited or not available. The generating set shall be rated for the power and duty cycle specified by the procuring activity (see 6.2).

3.11.1 Generating set portability. The generating set shall be skid mounted or trailer mounted as specified (see 6.2). When trailer mounted, the trailer shall be the manufacturer's standard conmercial trailer unless otherwise specified (see 6.2). The trailer shall be furnished with a ball-type hitch or steel lunette-type ring conforming to MS-51336, as specified (see 6.2).

3.12 Stud welding guns. The stud welding guns shall be of the portable, pistol type with a finger-actuated switch. The housing shall be made of high-Impact plastic or other non-conductive material with electrical insulating properties. Control and gun power cables shall enter the bottom of the handle or be secured to the bottom of the handle. The gun shall be capable of performing the required stud welding cycles when connected to the power sources specified herein. Each gun shall not overheat or wear prematurely when welding studs of the size specified herein when operated in an ambient temperature of 100 degrees Fahrenheit (F) without auxiliary or forced cooling.

3.12.1 Gun accessories. Gun grips, foot and leg assemblies, and chucks shall be provided as specified by the procuring activity (see 6.2). All gun accessories supplied shall fit the stud gun without requiring modification to the gun or accessory and shall weld all stud diameters required for the type and size system specified in 1.2.

3.12.2 Gun tools. Each stud gun shall be furnished with a complete set of tools required for its operation and maintenance.

3.13 Welding cables. The welding cables carrying the welding circuit current and the welding circuit ground shall be the manufacturer's standard resistant to oil, acid, alkali, and heat. The cables shall be capable of carrying the amperage necessary to weld all stud diameters required of the type and size of system specified in 1.2. The cables shall have quick disconnect

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connectors. Unless otherwise specified (see 6.2), type II systems shall be provided with not less than a 10-foot control cable, a 10-foot welding cable, and a 10-foot ground cable with clamp; and type VII systems shall be provided with not less than a 15-foot control cable, a 15-foot welding cable, and a 15 foot ground cable with clamp. Combination control and welding cables can be provided.

3.14 Gas shielding equipment. When specified for type VII systems (see 6.2), a gas flow meter and regulator to control the flow of shielding gas to the stud gun and to measure the cylinder pressure; shielding gas hose suitable for use with all gases used in the metal inert gas shielded and metal gas shielded welding process; a gas spark shield; and all necessary adaptors and connectors for shielded arc welding shall be provided. Additional gas shielding equipment required by the procuring activity shall be as specified and fully described (see 6.2).

3.15 Primary connections. Access for the primary power leads shall be provided through the rear or side panel of the enclosure. All primary leads shall be connected inside the enclosure.

3.16 Performance. Each stud welding system shall be capable of welding studs to meet the requirements in 3.16.1, 3.16.2, and AWS C5.4.

3.16.1 Weld visual inspection requirement. The fillet formed by stud welding using type VII systems (type II systems do not form a fillet) shall be a well-formed fillet, wetted to the base plate. All welds (except for aluminum welds) shall have a bluish cast with no evidence of undercutting, inclusions, porosity, cracks, or excessive spatter.

3.16.2 Weld bend requirement. Ten studs of each diameter welded in 4.7.2.3 shall be hammered over or bent using a bending tool 15 degrees for type II systems and 45 degrees for type VII systems or until the stud or base plate fractures. Failure shall occur in the stud or on the base plate or a plug of base metal shall tear out. There shall be no evidence of failure or cracking in the weld zone.

3.17 Standard equipment. All standard equipment normally provided with the manufacturer's commercial welding system shall be furnished.

3.18 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or system.

3.19 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.20 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

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3.20.1 Nameplate. A nameplate(s) shall be securely attached to each system. Unless otherwise specified (see 6.2), the nameplate(s) shall contain the information listed below. If the system is a special model, the model designation shall include the model number of the basic standard system and a Suffix identified in the manufacturer's permanent records.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.20.2 Weld cycle chart. Recommended welding times and currents for the stud diameter range applicable to the size furnished shall be permanently marked on a chart or plate near the weld timing and current controls on each type VII system. Weld diameters shall be given in inches and millimeters.

3.20.3 Diagram of connections. A diagram of electrical connections shall be attached to the inside of each system enclosure. The diagram shall be protected with transparent varnish, transparent plastic covering, or other transparent material.

3.21 Workmanship. Workmanship of the welding system shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance, All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The system shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance Inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7.2.1 and 4.7.2.2, and the inspection in 4.8. The system shall pass the examination, all tests, and the Inspection to be accepted.

4.6 Examination. Each item shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 50 degrees to 104 degrees F and 20 percent to 95 percent relative humidity.

4.7.2 Performance tests. Each system shall be capable of performing all tests specified herein as applicable to the type and size system furnished. At the option of the procuring activity, each system shall be subjected to tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2). All performance tests shall be completed without performing any maintenance on the system.

4.7.2.1 Operational test (type II and type VII). All circuits in the welding system shall be energized. Proper operation of all controls, motors, adjusting mechanisms, optional equipment, and accessories shall be verified without performing any actual welding.

4.7.2.2 Maximum current output test (type VII). The stud welding system shall be tested to determine that the maximum current output meets the requirements for the maximum current output for the power source given in Section 3 for the type and size system furnished.

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4.7.2.3 Welding test (type II and type VII). Each system shall be tested to ensure it meets the performance requirements specified herein. Round, low carbon steel studs (c-1008 to C-103O) of the smallest diameter in the applicable stud diameter range shall be welded to a low carbon steel base metal (C-1008 to C-103O) for 10 minutes at the welding rate required for the type and size furnished. The test shall be repeated using the largest stud diameter given in the stud diameter range. The base metal shall be not less than 1/3 the thickness of the diameter of the stud. All welds shall conform the requirements of 3.16.1 and 3.16.2.

4.7.2.4 Gas shielded weld test. When gas shielding equipment Is specified (see 3.14), the equipment shall be tested to ensure it functions properly and meets the performance requirements specified herein. Round, aluminum alloy (5356) studs of the largest diameter in the applicable stud diameter range shall be welded to an aluminum alloy base metal (1100, 3000, or 5000 series) for 10 minutes at the welding rate required for the type and size furnished. The base metal shall be not less than 1/3 the thickness of the diameter of the stud. Proper operation of the gas shielding equipment shall be determined. All welds shall pass the inspections in 3.16.1 and 3.16.2.

4.7.2.5 Capacitor discharge system safety devices (type II only). The operation of the automatic discharge safety device (see 3.4.4) shall be tested and observed. After 30 seconds, no indications of voltage shall be observed. The protective circuit preventing capacitor voltage discharge prior to securing the stud gun on the workpiece, when provided (see 3.4.4), shall also be tested and observed to determine proper operation.

4.7.2.6 Automatic failure protection test (type VII). An overload condition, excessive temperature rise, excessive weld cycle time, or component failure shall be simulated to test the automatic failure protection device. The protection device shall comply with 3.9.

4.7.2.7 Overtemperature protection test. Systems provided with forced air cooling (see 3.7) shall be tested for automatic excessive temperature rise failure protection. With the system operating in a thermally stabilized condition at rated load current and with enclosure panels in place, the fan motor shall be disconnected. The automatic failure protection device shall comply with the requirements of 3.9.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. The requirements for packaging shall be in accordance with MIL-M-45562.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The stud welding systems covered by this specification will be used to weld carbon steel, stainless steel, and aluminum alloys. Galvanized steel, copper, and brass can also be welded with special provisions. The stud welding systems will be manually loaded and operated in indoor and outdoor environments such as construction and fabrication sites, repair shops, shipyards, aboard ships, and aircraft and ground support equipment sites.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. Additional safety and health requirements, if required (see 3.2.3).

f. Painting, if different (see 3.3.4).

g. Specify drawn arc, initial gap, or initial contact type II capacitor discharge system, as required (see 3.4).

h. Input power requirements for type II size 1 system, if different (see 3.4.1).

i. Input power requirements for type II size 2 system, if different (see 3.4.2).

j. Input power requirements for type II size 3 system, if different (see 3.4.3).

k. Type II system protective circuit, if required (see 3.4.4).

1. Input power requirements for type VII, size 1 system, if different (see 3.5.1).

m. Input power requirements for type VII, size 2 system, if different (see 3.5.2).

n. Heavy duty gun on type VII, size 2 system, if required (see 3.5.2.1).

o. Dual guns on type VII, size 2 system, if required (see 3.5.2.2).

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p. Input power requirements for type VII, size 3 system, if different (see 3.5.3).

q. Dual guns on type VII, size 3 system, if required (see 3.5.3.1).

r. Input power requirements for type VII, size 4 system, if different (see 3.5.4).

s. Dual guns on type VII, size 4 system, if required (see 3.5.4.1).

t. Input power requirements for type VII, size 5 system, if different (see 3.5.5).

u. Dual guns on type VII, size 5 system, If required (see 3.5.5.1).

v. Input power requirements for type VII, size 6 system, if differnt (see 3.5.6).

w. Memory analyzer board on type VII system, if required (see 3.5.9).

x. Portability, if different (see 3.10).

y. AC power generating set, if required (see 3.11).

z. Generating set power and duty cycle requirements (see 3.11).

aa. Trailer or skid mounted generating set, as specified (see 3.11.1).

bb. Trailer, if different; specify ball-type hitch or steel lunette ring (see 3.11.1).

cc. Specify and fully describe gun accessories, if required (see 3.12.1).

dd. Specify cable length, if different (see 3.13).

ee. Specify gas shielding equipment, if required or if different (see 3.14).

ff. Specify and fully describe optional equipment, if required (see 3.18).

gg. Specify applicable repair parts, if required (see 3.19).

hh. Nameplate, if different (see 3.20.1).

ii. Inspection conditions, if different (see 4.3).

jj. First article inspection, if different (see 4.4).

kk. Quality conformance inspection, if different (see 4.5).

11. Performance test(s), if different (see 4.7.2).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
•	•	•	•
•	•	•	•

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (S1)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

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6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

MIL-W-23680D	MIL-W-23680E
Type I, Classes 1 and 2	Deleted (now covered by MIL-W-80110)
New	Type II, Size 1
Type II, Classes 1, 2, and 3	Type II, Size 2
New	Type II, Size 3
Type III, Class 1	Type VII, Size 1
Type IV, Class 1	Type VII, Size 1
New	Type VII, Size 2
Type IV, Class 2	Type VII, Size 3
New	Type VII, Size 4
Type IV, Class 3	Type VII, Size 5
New	Type VII, Size 6
Type V, Class 1	Type VII, Size 5
Type VI, Class 1	Type VII, Size 1
Type VI, Class 2	Type VII, Size 2
Type VI, Class 3	Type VII, Size 5
Type VI, Class 4	Type VII, Size 6

6.9 Safety and health requirements. Regulation 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new system into its future environment.

6,10 Electrical requirements. There are no existing non-Government or Government specifications or standards covering electrical requirements for stud welding systems.

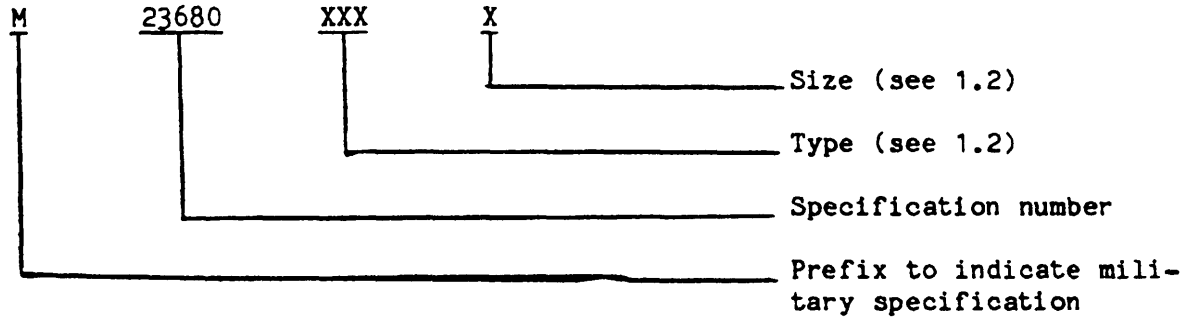
6.11 Training. Training required by the procuring activity should be provided as specified in the contract.

6.12 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

6.13 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

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6.14 Part or Identifying Number (PIN). The PIN to be used for systems acquired to this specification are created as follows:



(Note: The Roman numerals used to identify system "type" shall be changed to Arabic numbers when forming the PIN, i.e., "II" shall be written as "2".)

6.15 Subject term (keyword) listing.

Arc welding
Joining
Shielded welding
Torches
Unshielded welding

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy - YD
Air Force - 84

User activities:
Army - CE, ME
Navy - CG, OS

Preparing activity:
DLA - IP

Project (3431-0188)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-23680E	Stud Weld Sys, Int DC Power Sources & Control Units

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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